DX700 DATA SCIENCE CAPSTONE · MODULE C

IntentCX

Routing customers by conversion intent

Results of a digital marketing capstone: Section 6, with the tuning work from Section 5.4

Lauren Chagaris · Boston University, Online MS in Data Science, October 2026

What IntentCX does

1

Score

A classifier rates each customer's likelihood to convert.

2

Route

High-intent customers go to premium agents. Low-intent customers go to the Frugal Gateway, a cost-optimized path.

3

Save

Fewer costly interactions with customers unlikely to convert lowers cost per conversion and AI inference load.

8,000
customers in the dataset

21
modeling features

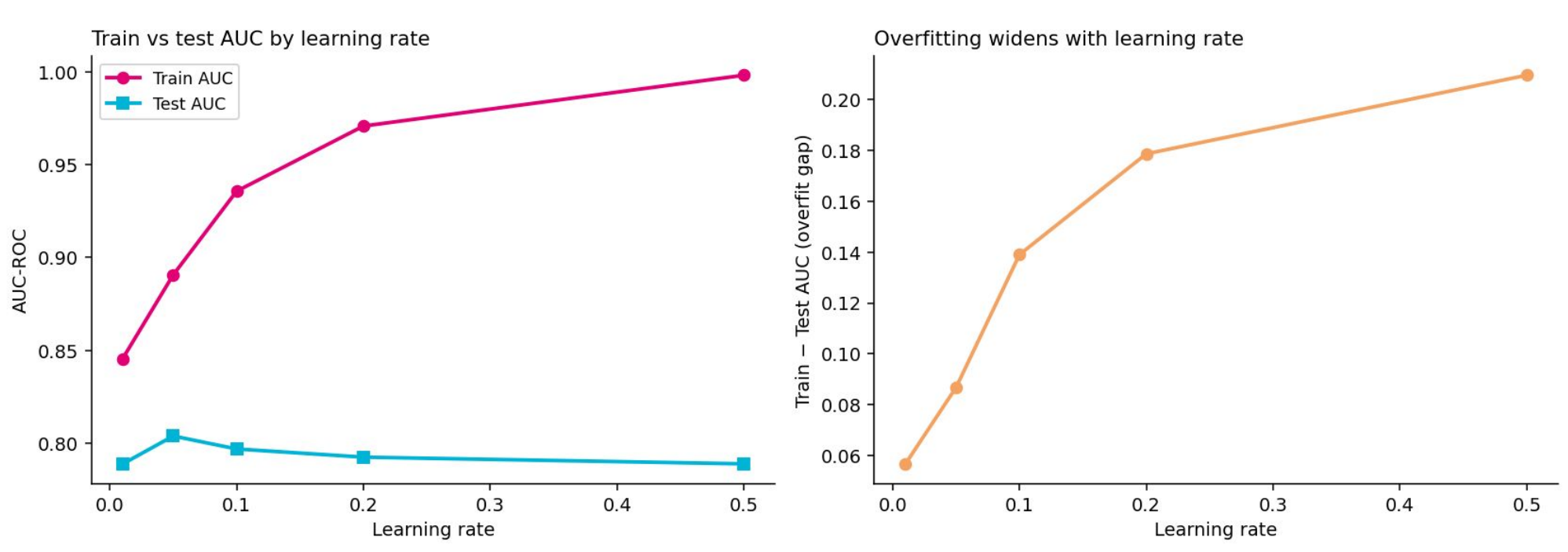
87.65%
of customers converted

Gradient Boosting performs best

Model	Held-out AUC-ROC	Notes
Logistic regression	0.7708	Linear baseline
Linear SVM	0.7654	Standardized features
K-nearest neighbors	0.7680	Manhattan, k = 101
Random Forest	0.7976	Cross-validated AUC 0.8215
Gradient Boosting	**0.8066**	**Cross-validated AUC 0.8280, SD 0.0244**

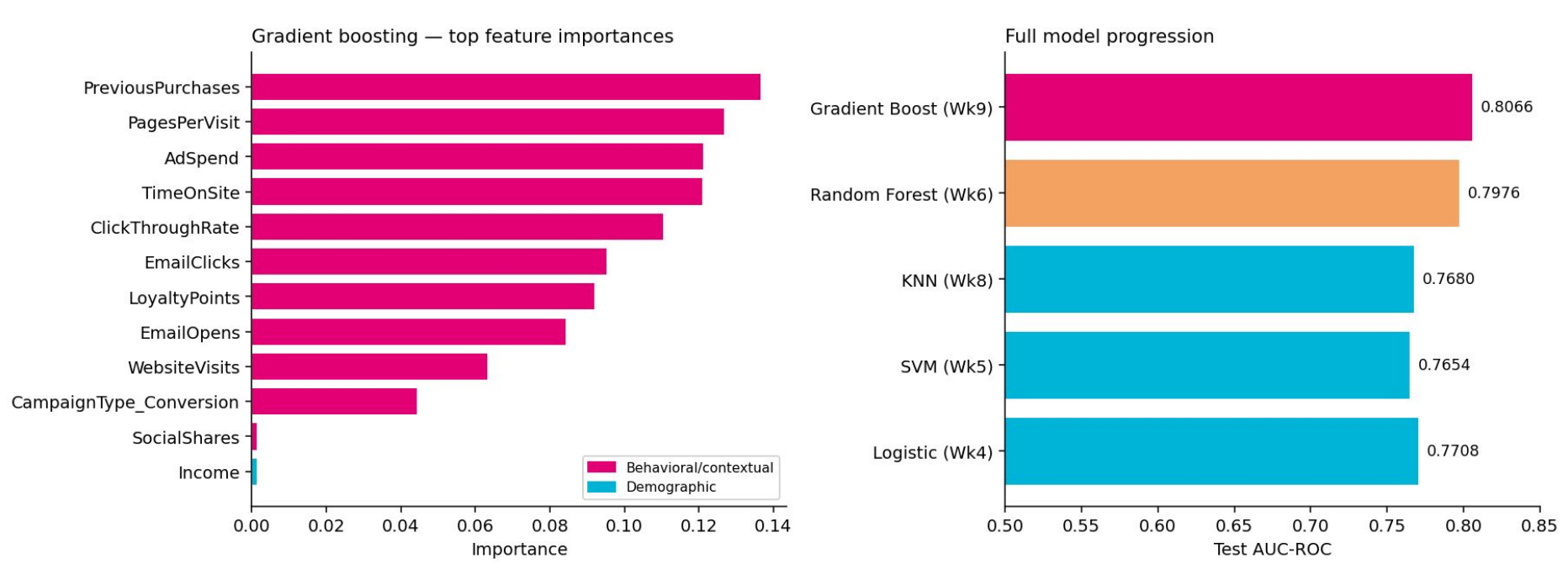
The regression family (OLS, Ridge, Lasso, ElasticNet, PCR, PLSR) reached an R-squared near 0 on the campaign-level target. Boosting beats the logistic baseline by about 0.036 AUC.

Learning rate controls overfitting



From 0.01 to 0.5, train AUC climbs from 0.85 to 0.998 while test AUC stays near 0.79. Final model: learning rate 0.1, depth 2, 100 trees.

Behavior drives the score



Previous purchases, pages per visit, ad spend, time on site and click-through rate lead. Demographics add almost nothing, so a deployed score can omit sensitive attributes.

Dilemma 1. Class imbalance

With 87.65% of customers converting, accuracy is misleading, so AUC-ROC is the primary metric.

99.4%

recall: 1,394 of 1,402 converters stay on the premium path

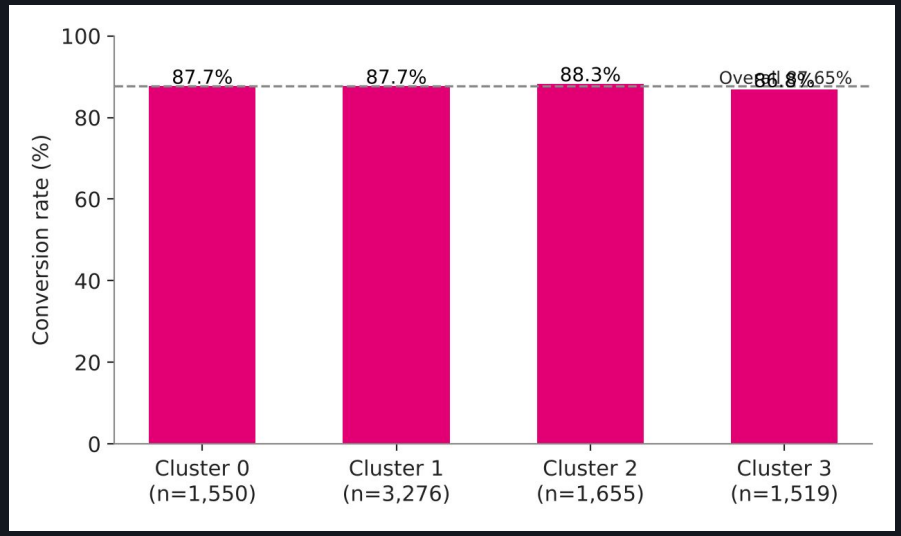
24.7%

specificity: only 49 of 198 non-converters are caught

Question: How should the routing threshold be set when the dataset has no cost data?

Dilemma 2. Reporting a null result

Method	Silhouette	Conversion rate by cluster
K-Means (k = 4)	0.1016	86.8% to 88.3%
DBSCAN	-0.1118	89.4% main cluster vs 83.2% noise
Hierarchical, Ward (k = 3)	0.0810	87.3% to 88.2%



Question: Do three clustering methods that found nothing strengthen the report, or distract from it?

Dilemma 3. How far do results travel?

1

source, with no documented collection method

≈ 0

correlations and uniform feature spreads suggest synthetic data

0.6136

AUC when a similar approach was tested on a business-to-business dataset

Question: How confident should the recommendations sound, and what is the most honest way to say so?

Recommendations

1

Pilot in shadow mode

Score customers with Gradient Boosting on behavioral and campaign features only, without changing routing, and compare predicted intent with actual conversion.

2

Tune the threshold

Price a premium interaction and a missed conversion with the client's own costs, which also produces the profit curve this report lacks.

3

Validate independently

Test on at least one other dataset, collect cost and revenue data, and audit routing outcomes across customer groups.

Three questions for you

1. How would you set a routing threshold when the data has no costs?
2. Does a null clustering result strengthen the report or distract from it?
3. How confident should conclusions sound with one probably synthetic dataset?

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